Visibility Is Not Detection: Auditing Measurement Boundaries in Staged and Evidence-Ranking Evaluation

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Abstract:
Final-answer scores can hide whether an evaluation observed a defect, consumed the observation, or treats the outcome as harmless. We audit exactly two constructions. In a staged document evaluator, 96/118 header injections are visible, while the OCR-like family is unobserved on the frozen surface (0/13) and harmfulness remains contract-relative. Flat and layered scores are affine/rank-equivalent, a clean-only panel cannot identify the deployment threshold, and explicit instrumentation explains perfect stage separation. A nested FinQA probe exposes a measurement-object mismatch. In FinRank, tie policy moves absolute accuracy but preserves the hard-versus-random direction; post-hoc text_sha1 correction finds 442 overlapping hard-negative occurrences across 305 records and gold loss in 33 records. The bounded contribution separates visibility, detection, harmfulness, score sensitivity, decision stability and reproduction gap, and proposes a review checklist rather than a universal theory.

1 INTRODUCTION

Staged document pipelines are usually reduced to final-answer correctness. That score cannot distinguish an observed defect from silent absorption or an unobservable defect, so it can hide whether the evaluation surface is capable of seeing the failures it claims to assess. We separate six objects: visibility (instrument observation), detection (predicate consumption under a threshold), harmfulness (outcome change under an equivalence contract), score sensitivity (absolute metric movement), decision stability (survival of the supported decision), and reproduction gap (artifact agreement under a stated environment). These are audit bookkeeping terms, not a causal theory. Confounding them turns score movement into protocol sensitivity, non-response into harm, and metric drift into decision reversal. Figure 1 shows the chain. This paper audits two evaluation constructions. Case 1 is a frozen staged document evaluator with controlled header perturbations and a stage-aligned instrumentation study. It exposes family-dependent visibility, affine/rank score equivalence, a non-identifiable clean calibration point and an instrumentation attribution boundary. Its FinQA program-perturbation probe is nested inside Case 1 and tests only whether the perturbation and detection predicate refer to causally aligned measurement objects. Case 2 is FinRank, a public provenance-sensitive evidence-ranking benchmark. A source-informed implementation audit examines exact tie handling, first-occurrence text/source mapping, random-negative construction and numerical reproduction. FinRank v1 tie metrics and thresholds are preregistered and unchanged; its corrected source-identity branch is explicitly post-hoc v2. The FinRank hard-

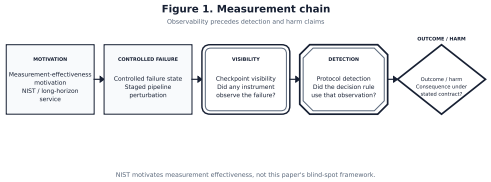


Figure 1: From injected failure to visibility, protocol detection and contract-relative outcome harm. NIST motivates measurement effectiveness but not this paper's framework.

ness ordering remains stable under alternative tie policies, while absolute pairwise values and some ranking metrics are contract-sensitive. The same case also narrows a reproducibility claim: in-record and pairwise values reproduce exactly in the audited environment, but global and metadata-filtered values do not reproduce at raw precision and the cause is unresolved. Our contributions are:

1. A measurement-object separation for visibility, detection and contract-relative harmfulness in the staged evaluator, with deterministic evidence for score-structure and instrumentation identifiability.
2. A bounded FinRank evidence-ranking audit showing that tie policy and identity construction can move absolute metrics without changing the qualitative hard-versus-random decision, while keeping the post-hoc v2 identity correction and residual reproduction gap visible.
3. A proposed six-step review checklist for evaluator authors: bind mutation, observation, score, ranking and reported claim to the same measurement object; report tie and identity contracts; and state the numerical tolerance and environment required for reproducibility. This checklist is not a validated universal method.

The accounting is intentionally narrow: there are exactly two counted cases, with FinQA nested inside Case 1. MA-CERD and adaptive revalidation are outside Paper E and contribute no case or result. The audit does not establish a universal method, a detector ranking, or a general failure rate. Its external value is narrower but concrete: it changes what should be reported and which comparisons can be interpreted before further experiments are commissioned.

Table 1: Interpretation corrections exposed by the audit.

Naive reading	Audit evidence	Correct reading
Score gap implies better ranking.	Flat and layered scores are affine/rank-equivalent; the gap is threshold placement.	Report score algebra and threshold identifiability.
Non-trigger implies harmful blindness.	OCR-like is 0/13 visible; an existing judge calls the cases equivalent.	Non-response is an observation boundary, not harm.
Perfect localization comes from hierarchy.	Direct stage leaves match the full vector; non-direct leaves only partially separate.	Explicit instrumentation explains this construction.
Metric movement flips the decision.	Absolute values move, but the hardness gap preserves direction.	Separate score sensitivity from decision stability.
A deterministic rerun reproduces the benchmark.	Some metrics reproduce; global and metadata values drift.	Bind claims to environment, tolerance and metric scope.

2 RELATED WORK

2.1 Measurement validity and construct validity

Meta-evaluation studies whether metrics separate errors from acceptable variants or local contextual corrections (Amrhein et al., 2022; Amrhein et al., 2023; Moghe et al., 2024; Deviyani and Diaz, 2025); rankings also move with protocol axes, prompts, seeds and target properties (Kendiukhov, 2026; Du et al., 2026; Ross et al., 2024). LLM-judge work reports position, verbosity, self-preference and perturbation effects (Li et al., 2025a; Li et al., 2025b), while NIST asks for meaningful measurement and demonstrated construct validity (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024). We ask whether the observation surface can see a controlled defect before interpreting metric differences. Caban and Bean et al. already provide broad validity frameworks and benchmark checklists (Caban, 2026; Bean et al., 2025). FinRank separates retrieval, reranking and hard-negative discrimination on a curated passage pool (Mansouri et al., 2026). Our increment is narrower: bind observation surface, score contract, tie order and source identity to the object that a concrete evaluation claims to measure.

2.2 Testing, oracles, and diagnosability

Mutation and metamorphic testing inject controlled changes (Cho et al., 2025); fault reasoning separates reachability, state change and propagation. Process-level evaluation observes intermediate behavior (Fan et al., 2026; Chung et al., 2026). We decompose non-detection into visibility, predicate and outcome mechanisms and audit the checkpoint instrument itself.

2.3 Empirical software evaluation and reproducibility audits

RENE work similarly exposes measurement limits: Python-version identification degrades on realistic files (Gerhold et al., 2024); mining tools can disagree sharply on simple repository metrics (Hoess et al., 2025); test code alone is weak evidence for flakiness (Berndt et al., 2026); and a replication reverses part of the original symbolic-execution conclusion (Kummita et al., 2026). Our scope is exactly two constructions with explicit construct boundaries.

3 METHOD

The conceptual core is a three-way audit of visibility, detection, and harmfulness; the hierarchical checkpoint forest is the evidence instrument that makes the audit reproducible.

3.1 Evidence instrument: checkpoint forest

We model the pipeline as two checkpoint trees. The parsing tree covers document ingestion, table structure, field mapping, and arithmetic reconciliation. The QA tree covers retrieval grounding, faithfulness, numeric correctness, citation quality, and appropriate refusal. Together the trees contain 12 scored leaves. The twelve scored nodes are five in the parsing tree (document status, document throughput, table regression, field hit-rate, reconciliation equations) and seven in the QA tree (retrieval contract, faithfulness judge, numeric contract, number grounding, citation judge, refusal judge, refusal contract), per the frozen checkpoint-tree configuration v0.1.0 (listed in the artifact manifest). The visibility audit exercises the parsing-tree leaves whose evidence is the mapping and regression artifacts; under the frozen drop rule (drop = (H_orig - H_inj) x 10, threshold 2.0) only the field hit-rate leaf responds to header drift, while QA-tree leaves keep their frozen evidence and the QA arm is reported separately (12 calls to the same judge). Additional localization value therefore remains unestablished in the current instantiation. In Table 2, the label Pruned root drop denotes exactly this operation: the parsing root is collapsed to its single input-responsive leaf (the field hit-rate leaf) and the drop rule above is applied there. Each leaf receives a score in [0,10]. Contractible properties use deterministic checks; semantic properties use a 0–5 judge score mapped to [0,10]. Sibling weights are normalized locally before bottom-up aggregation. Missing evidence is explicit: the parsing tree assigns 0, whereas the QA tree assigns a neutral 5. Hinge leaves affect the stability verdict, not the missing-evidence score. Every leaf stores its evidence and degradation reason. For node n with children C(n), the recursive score is

$$S(n) = \sum_{c \in C(n)} \frac{w(c)}{\sum_{j \in C(n)} w(j)} S(c).$$

For a leaf, S(n) is the mapped contract or judge score. A tree is stable only when at least 90% of its leaves exceed the selected threshold and every hinge leaf scores at least 7. The implementation is deterministic once the evidence artifacts are fixed.

3.2 Separating visibility, detection, and harmfulness

Visibility is defined relative to the selected checkpoint evidence surface, whereas protocol detection may use a different downstream signal. The primary conceptual structure is the following visibility-by-detection matrix: Harmfulness is orthogonal to this matrix and is annotated separately as harmful, equivalent, or unresolved under the selected equivalence criterion. For an injected defect d, visibility and protocol detection are separate observations:

Table 2: Visibility-by-detection measurement matrix.

Checkpoint evidence	Protocol detects	Protocol does not detect
Visible	Detected visible defect	Visible but undetected
Not visible	Detection through a different signal or predicate	Silently absorbed or unobservable

- Detected: the selected protocol satisfies its detection predicate, which may test a mapping change or a score threshold.
- Visible but not detected: the checkpoint exposes a difference, but the selected protocol does not satisfy its detection predicate.
- Silently absorbed: normalization or matching preserves the canonical correspondence, so that the checkpoint does not expose the perturbation.

Harmfulness is a separate, equivalence-dependent judgment, not a mutually exclusive visibility outcome: canonical-form preservation and reader-referent recoverability need not agree. The defect visibility profile reports visibility separately for each family. This preserves information lost by a single mutation score and distinguishes a missing checkpoint from a genuinely robust stage.

3.3 Evaluating the evaluator

We compare four protocols on frozen evidence: end-to-end roots, a flat leaf average, hierarchical expert weights, and hierarchical searched weights. We measure table-level detection, injection-level detection, mean separation between clean and perturbed states, worst-family behavior, localization, and evaluation cost. Because raw thresholds can manufacture differences, we separately audit (i) structural score/ranking equivalence and (ii) whether a matched-FPR operating point is identifiable from clean scores alone. The audit further varies the injection seed, visibility threshold, matching algorithm, and scoring normalization. A semantic-judge arm is evaluated separately and never substituted for deterministic evidence when a contract exists.

3.4 Stage-aligned instrumentation and the nested FinQA boundary probe

The evaluation-track instrument is a two-root checkpoint forest whose twelve leaves map deterministic or judge-derived evidence to a 0–10 score (Section 3.1). To test whether stage-aligned instrumentation carries stage information, we preregister a separate controlled study on the same corpus. Each stage contract perturbs one operation: a header token is replaced (mapping stage M), required source blocks are removed from the retrieval bundle (retrieval stage R), or one numeric operand is broken while labels, schema and identity are preserved (numeric stage N). Every cluster contributes three deterministic injections, and each injection is scored on the same twelve-leaf surface, producing a binary response vector; 24 clusters per stage give 72 clusters and 216 injections. The study is frozen before results (hash receipt) and evaluated leave-one-cluster-out. The same document defines the protocol's deliberate boundary: response predicates are written from the stage contracts, so the construction is explicitly instrumented and its separability must be interpreted within that construction. FinQA is a nested boundary probe inside Case 1, not a separate counted case. Its program-perturbation corpus acts on a step's operator or argument fields while the downstream result fields are not recomputed; the observation objects are visibility (does the injected change appear in the observed source fields), detection (does the frozen detection predicate move), and outcome (does any downstream outcome field change). The execution contract binds artifact hashes before scoring, and no threshold or convention search is permitted. Section 4.5 states the execution boundary; Section 5.6 reports what happened and what it does and does not establish.

3.5 External evidence-ranking audit

Case 2 is FinRank at upstream revision 916dc27e77b90e294ea1b3a2cc5b0abcf07392d1, with arXiv record 2608.07400v1, 1,185 released records and the official deterministic baseline implementation. The repository license is CC BY-NC 4.0. The audit follows the complete chain: question and source identity, candidate construction, scoring, global and in-record ranking, pairwise comparison, random-negative sampling, metadata filtering and published claims. The source code exposes two known, source-informed checkpoints before application: pairwise accuracy counts exact ties for the gold (score(gold) >= score(negative)), and pooled-corpus passage text is deduplicated in first-occurrence order so the first occurrence supplies metadata. These probes are not presented as prospective discoveries. FinRank v1 fixed the question, tie metrics and materiality thresholds before pair-level outcomes and remains the preregistered application, but its raw-exact identity rule was wrong for FinRank's canonical contamination identity. V2 is a post-hoc deterministic correction to trimmed text_sha1 identity and a full reanalysis; it is not a second blind application or a wholly frozen application. The v1 tie metrics

and thresholds are unchanged in v2. The audit compares ties-positive, strict-win and half-credit pairwise policies; gold-first and negative-first ordering for exact ties inside each record; forward and reverse corpus ordering for global ranking ties; the seed-42 random-negative draw; and source-identity loss under the metadata filter. Reproducibility is checked against the released result JSON under the declared dependency versions and a second available environment. The application is deterministic and source-informed, not blind.

4 EXPERIMENTAL SETUP

4.1 Corpus and perturbations

The study uses a frozen public tabular base and contains no client data. It includes 95 tables and 118 synthetic injections across six header-drift families. Schema and header matching are the perturbation carrier here; the object of study is the evaluation protocol's observation boundary. Each injected record preserves its source table, family, seed, changed field, and expected observation surface. Concretely, the base is the frozen drift-v0 table set derived from public TAT-QA tables (listed in the artifact manifest by content hash); the corpus was frozen before this study's data gates and no new TAT-QA use is introduced. The perturbations are the fully synthetic injection map (listed in the artifact manifest by content hash; deterministic, seed 42, ratio 0.3). The matcher used in the audited pipeline is the greedy header matcher (listed in the artifact manifest); the cross-matcher audit adds greedy-normalized, Hungarian, and Hungarian-normalized configurations. Semantic leaves are scored by the judge pipeline (listed in the artifact manifest), i.e. Qwen3.6-35B-A3B (Q4_K_M GGUF) served locally, with 0–5 dimension scores mapped to [0,10]. The primary threshold grid is

$$\Gamma = \{0.2, 0.3, 0.5, 0.7, 0.9\}.$$

The OCR-like family contains 13 single-token, lowercase Levenshtein-distance-one substitutions with unchanged column values. These properties are mechanically recomputed before interpretation.

4.2 Baselines and controls

The protocol baselines are end-to-end scoring, flat leaf scoring, and hierarchical scoring with expert weights. Weight-search controls include equal weights, expert weights, 30 random trials, greedy coordinate ascent, particle-swarm optimization (PSO), and a minimax objective. Assignment controls compare greedy matching with Hungarian global assignment and include a positive solver test.

4.3 Statistical discipline

We freeze inputs and decision criteria before reading each result. Per-seed confidence intervals use the injected source-table units in that seed (percentile bootstrap with seed 20260830). For the pooled three-seed sensitivity analysis, repeated source-table UIDs are treated as clusters; the table-run ratio is retained as a descriptive point estimate, while uncertainty is recomputed by source-table UID-cluster bootstrap. With only three injection seeds, seed generalization is summarized by the observed range and leave-one-seed-out values rather than a population-level seed interval. For the clean-only matched-FPR audit, threshold selection may use clean scores only; injected and harmful outcomes are prohibited from calibration. Because the frozen clean supports are degenerate, the registered deployment operating point is left unresolved rather than forced into an empirical equality. The 2,382 clean records are the usable (table_uid, column) rows of the non-injected source tables in the frozen clean-input set, with fold assignment = sorted(table UID) index modulo 5. Null and adverse findings are retained rather than relabeled as improvements.

4.4 Multi-leaf stage-aligned protocol

The stage-aligned study uses the same frozen tabular base and the same twelve-leaf surface. Stage contracts and injection generators are deterministic; clusters are selected by an outcome-blind hash rule (seed frozen before results), and the pre-result freeze receipt binds the corpus, the generator, the runner and the environment by hash. Response predicates are contract-derived: for M the directly targeted parse leaves move; for R the retrieval leaf moves; for N the reconciliation leaf moves. Baselines are the metric-specific chance top-1 (1/3) and majority macro-F1 (1/6); uncer-

tainty uses leave-one-cluster-out evaluation with 10,000 cluster-bootstrap resamples and a 10,000-replicate permutation null, both seeds frozen. The ablation re-scores the same rows with (i) the full twelve-leaf vector, (ii) the fixed three-direct-leaf codebook, (iii) non-direct leaves only, and (iv) a single any-response scalar.

4.5 Second-dataflow execution boundary

The nested FinQA boundary probe executes a frozen step-perturbation corpus under a hash-bound binding contract: 89 rows over 30 clusters, with per-row runtime bindings and a full trace map verified before scoring. The perturbation writes operator/argument fields; the observation vector records visibility, detection and outcome per row. Because downstream result fields are not recomputed from the mutated program semantics by construction, a zero detection count in this execution is not an estimate of detection ability; Section 5.6 states the resulting measurement-object boundary and the alternatives the evidence does not exclude. No threshold search, convention selection, or source-operator substitution is permitted in this probe. It is nested inside Case 1 and does not increase the case count.

4.6 FinRank external audit setup

FinRank contains manually authored questions, reference answers, gold passages and curated hard negatives over 22 companies. The released global pool contains 5,230 unique raw passage texts. The audit uses only the released JSONL, repository code, split/random seed and result files. TF-IDF and BM25 are regenerated from the official script with its released parameters; neural baselines are outside this audit. The primary proposition was fixed before pair-level outcomes: the tie-positive policy and first-occurrence identity mapping may change absolute metric interpretation, while the paper's narrow curated-pool contract may keep the hard-versus-random conclusion stable. V1 materiality thresholds are 1.0 percentage point for pairwise values or hardness-gap movement and 0.5 point for global tie-order movement, and they remain unchanged after the post-hoc v2 identity correction. The outcome classes are interpretation changed, narrower contract stable, or no substantive impact.

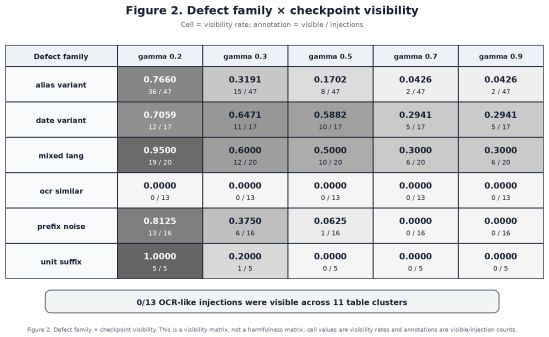


Figure 2: Defect family × checkpoint visibility. Cell values are visibility rates at each threshold, annotated with visible/injection counts; the OCR-like family is 0/13 across 11 clusters at every tested threshold. This is a visibility matrix, not a harmfulness matrix.

5 RESULTS

5.1 Visibility depends on defect family

Checkpoint-level analysis makes 96/118 injections visible, but the aggregate hides a large family effect. 0/13 OCR-like injections were visible across 11 table clusters under all five thresholds; all 13 were invisible under this observation criterion. Other families range from 0.81 to 1.00 visibility at the primary setting. For uncertainty, the injection-level reference estimand is 0/13 with an exact binomial 95% interval of [0, 24.71%] under an independent-injection assumption; the distinct table-level event estimand is 0/11 with an exact binomial interval of [0, 28.49%]. The latter is not a clustered confidence interval for injection-level visibility. Figure 2 reports the full family-by-threshold visibility profile. The family unobserved on the frozen scored surface remains the same under four edit-distance matcher configurations. However, the ordering of visible families is not invariant: changing the score normalization causes five class flips, all within the date-variation family, whereas switching from greedy to global assignment causes no flips. Thus, the existence of a family unobserved on the frozen scored surface is stable within the tested matcher family, but the full ranking is protocol-dependent. The harmfulness label is also criterion-dependent, and our record contains two comparison kinds that should not be pooled. In the first kind, seven paired judge-versus-standard comparisons yield four disagreements; the two standards differ in the criterion they apply, with the standard asking whether the canonical form and canonical truth survive and the judge asking whether a reader can recover the same referent. In the second

kind, a single visibility-versus-equivalence comparison contrasts deterministic channel evidence with standard equivalence, and that case carries no judge call. Because the two kinds measure different objects, we keep the raw cases and report neither a pooled disagreement rate nor a mechanism that would exclude case-level noise. In particular, the existing single judge, applying its table-context reader-referent contract, classified all 13 OCR-like perturbations as referentially equivalent. That supports benign absorption as an alternative interpretation of the non-response; it is not independent ground truth, and non-response alone does not establish harmful false negatives. We therefore report that this family is unobserved on the frozen scored surface, while treating whether that absence is harmful as a boundary question.

5.2 The raw hierarchy gap is a threshold-placement artifact

The raw detection counts are not evidence of protocol discriminability. Table 2 compares three transferable protocols under different decision predicates: end-to-end detection tests whether the final mapping text changes, whereas flat and hierarchical scoring use frozen score-drop thresholds. A changed source-header string can therefore count as end-to-end detection even when the canonical match is preserved, explaining why 112/118 exceeds the 96/118 visibility count. Table-level counts apply all assigned injections together, whereas injection-level counts apply one perturbation at a time; the 20 detected tables and 13 individually detected injections are not nested counts. At the frozen threshold, hierarchical scoring is not a more sensitive detector. The raw flat-hierarchical gap does not establish inferior ranking quality. In the current single-responsive-leaf construction, the unrounded hierarchical contrastive score is a positive rescaling of the flat contrastive score, with fitted slope approximately 0.375, and the recorded scores retain identical ranks; corresponding decisions, and by implication their ROC geometry, therefore coincide, and corresponding thresholds satisfy theta_H = 0.375 theta_F. This is a structural equivalence of the two protocols' decisions, not a newly calibrated performance estimate; the diagnosis is deliberately narrow: it explains why the raw protocol gap cannot be read as a ranking difference, and it is not offered as a general algebraic result about hierarchical evaluation. Separately, the registered deployment operating point remains unresolved: the derived clean-only reanalysis contains 2,382 usable clean records whose score supports are exactly {0}, so every positive threshold has an observed FPR of zero and the clean panel cannot select a unique operating point. That calibration degeneracy does not negate the structural equivalence, and it is a statement about the current clean-only procedure, not a general claim that matched-FPR comparisons cannot be made; the registered strict predicate FPR = alpha and the tolerance predicate FPR <= alpha also remain distinct. Selecting a threshold by harmful TPR would violate the clean-only contract. Historical fixed-threshold sensitivity evidence remains in the dated historical governance record, but the old [0,0] interval is not a clean-calibrated confirmatory result. When more than one leaf varies with the input, neither affine nor ranking equivalence is guaranteed. The 118 injections close at the checkpoint level as 96 visible plus 22 silently absorbed; by family (injections / visible / absorbed): alias variant 47/43/4, date variant 17/15/2, mixed language 20/20/0, OCR-like 13/0/13, prefix noise 16/13/3, unit suffix 5/5/0 (frozen artifact listed in the evidence ledger). The protocol-level counts in Table 2 are measured on different signals and are therefore not a partition of the 96. We label the affine-equivalence observation a degenerate-case result: it fixes the interpretation of the raw gap under the current single-varying-leaf construction and does not establish affine equivalence of the protocols in general.

Table 3: Protocol-level detection under different decision predicates.

Protocol	Detection signal	Tables detected	Injections detected
End-to-end	Final mapping change	95/95 (1.0000)	112/118 (0.9492)
Flat	Field-hit-rate drop	75/95 (0.7895; 95% CI [0.6947, 0.8737])	85/118 (0.7203)
Hierarchical expert	Pruned root drop	20/95 (0.2105; 95% CI [0.1263, 0.2947])	13/118 (0.1102)

5.3 Sensitivity and reproducibility

The identity of the family unobserved on the frozen scored surface is stable across the threshold grid. The absolute between-family range shrinks at high thresholds, as expected when one changed header contributes at most 1/n for a table with n canonical headers. The literal preregistered range criterion fails at 0.7 and 0.9, and this failure is reported. Across three injection seeds, flat table-detection rates are $75/95 = 0.7895$, $76/97 = 0.7835$, and $68/90 = 0.7556$. The denominator-weighted table-run estimate is $219/282 = 0.7766$, but these runs map to only 119 unique source-table UIDs: 29 appear in one seed, 17 in two, and 73 in all three. A post-hoc deterministic source-table UID-cluster bootstrap over 20,000 resamples gives a 95% interval of [0.7117, 0.8364] for the table-run ratio. Equal-weighting the source-table UIDs gives a sensitivity estimate of 0.7535 with interval [0.6835, 0.8207]. The observed seed spread is 0.0339; leave-one-seed-out rates are 0.7701, 0.7730 and 0.7865. With only three seeds, we report these sensitivity values rather than a population-level seed interval. The historical naive interval [0.7270, 0.8227] is retained only in the dated reanalysis record and is not presented as source-table uncertainty. The seed-42 run reproduces the frozen artifact byte-for-byte. Judge calibration covers 120 leaf-state cells; this calibration window is distinct from the 28-case semantic boundary analysis reported in Section 6. In the control-state comparison, rejudged summaries of historical 0–5 judgments differ from the frozen dimension means by at most 0.5, meeting the preregistered consistency expectation. This checks reproduction of aggregate dimension scores, not agreement between judges re-evaluating the same original answers. On the derived 0–10 root scale, control roots differ by at most 0.22, while individual leaves can move by as much as 7.0. Agreement is therefore an aggregate observation, not evidence of cell-level interchangeability.

5.4 Weight search produces an informative null

PSO and greedy coordinate ascent reach the same mean separation of 1.432574. PSO repeats the result on all five seeds but requires 7,650 evaluations, compared with 67 for greedy search. A minimax variant improves worst-case separation to 0.35714 while reducing mean separation to 1.25605. We therefore do not claim PSO superiority. The result instead shows that objective choice matters more than optimizer sophistication in the present search space.

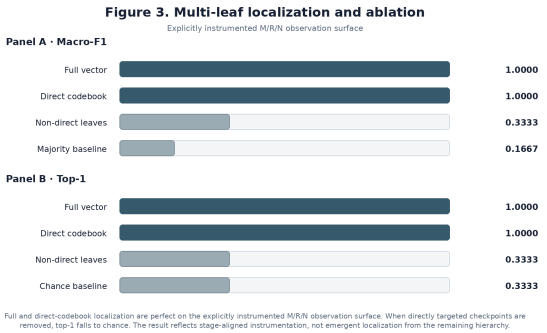


Figure 3: Multi-leaf localization and ablation. Perfect full-vector and direct-codebook bars are construct-sensitive on the explicitly instrumented controlled M/R/N observation surface. After directly targeted checkpoints are removed, non-direct top-1 equals chance (0.3333); under the frozen tie-as-incorrect rule these leaves still separate mapping from the retrieval and numeric stages without separating retrieval from numeric. Non-direct macro-F1 equals the uniform-class macro-F1 baseline (0.3333); the majority-class macro-F1 reference (0.1667) is plotted. The result reflects stage-aligned instrumentation, not emergent localization from the remaining hierarchy.

5.5 Stage-aligned instrumentation: separable signatures and their boundary

The preregistered stage-aligned study separated the three controlled stages perfectly on the explicitly instrumented surface: leave-one-cluster-out macro-F1 and top-1 accuracy are 1.000 with cluster-bootstrap 95% intervals [1.0, 1.0] and permutation $p = 1/10001 = 1e\text{-}4$, with zero ties and zero errors, against metric-specific baselines (chance top-1 1/3; majority macro-F1 1/6). The per-stage response profiles are disjoint within the construction: M moves the field-hit and table-regression leaves, R moves the source-required leaf, and N moves the reconciliation leaf. Figure 3 reports the localization and ablation outcome. That positive result carries its boundary in the same rows. Re-scoring the identical 216 injections with the fixed three-direct-leaf codebook also yields macro-F1 and top-1 of 1.000; the non-direct leaves alone fall to 0.3333 with a tie rate of 0.6667, which is the three-class chance level and the uniform-class macro-F1 baseline; under the frozen tie-as-incorrect rule they distinguish the mapping stage from the retrieval and numeric

stages but do not distinguish retrieval from numeric, so they carry partial rather than no stage information. A single any-response scalar collapses to 0.0 with ties everywhere. Eight of the twelve leaf columns are identically zero over all injections. Perfect separability therefore reflects explicit stage-aligned instrumentation in a contract-derived construction, not emergent localization from the remaining hierarchy, and the positive result is reported together with this ablation. Per-stage estimates are descriptive at this sample size; the population is the frozen controlled injection set, not a natural-failure distribution.

5.6 Nested Case 1 boundary probe: measurement-object mismatch

The nested FinQA probe produced V = 89/89, D = 0/89 and O = 0/89: every injected change was observable on the mutated source fields, the frozen detection predicate never moved, and no downstream outcome field changed. The detection predicate here is defined and executable, and these counts are reproducible; what the perturbation does not do is regenerate the downstream result fields that the predicate reads. The zero detection count therefore describes this carrier construction rather than an estimated detection ability, and the mismatch is one of measurement objects, not of predicate definition. The formal result class is a mixed second-flow outcome; the accepted interpretation is a measurement-object mismatch, recorded in a separate autopsy. This is a boundary probe inside Case 1, not a third case. We therefore treat causal alignment of measurement objects as a design requirement for future transfers, not as a validated law. Scorer non-portability, threshold non-transfer and corpus shift are not excluded by this evidence, and no transfer, blindness or semantic-correctness claim is made from this probe.

5.7 Case 2: stable hardness decision, sensitive score contract

The compact FinRank result separates score sensitivity from decision stability. Values are hard/random pairwise accuracy and the random-minus-hard gap in percentage points: Exact ties occur in both deterministic score matrices. TF-IDF has 91 hard-pair and 135 random-pair ties; BM25 has 83 and 125. Changing ties-positive to strict-win moves random-negative accuracy by 1.084 points for TF-IDF and 1.029 for BM25, crossing the preregistered 1.0-point materiality threshold. This is score sensitivity. By contrast, the hard-versus-random gap moves by only 0.359 and 0.343 points and retains its direction under strict-win and half-credit scoring. This is decision stability for the narrow qualitative hardness claim. Five of 11,790 seed-42 random draws equal a curated hard negative, too few to explain the gap. Table values are independently rounded to three decimals from full-precision results. Consequently, subtracting the displayed TF-IDF strict-win cells gives 13.820, while the full-precision gap 13.820514366083984 rounds to the reported 13.821. Tie order is a separate sensitivity probe. Inside each record, concatenating gold before negatives makes exact ties gold-first. Reversing tie priority lowers MRR by 0.101-0.108 points and nDCG@5 by 0.572-0.573 points. Forward-versus-reverse corpus order changes top-level global ranking metrics by at most 0.287 point, below the preregistered 0.5-point threshold. Absolute scores and small ranking differences are therefore contract-sensitive, while the qualitative ordering remains stable under the tested alternatives. The identity audit is explicitly post-hoc v2 after the wrong-identity v1 application. These are corrections to this audit's identity implementation, not corrections to FinRank itself. The canonical text_sha1 branch reproduces the source-informed, author-disclosed target quantities of 442 hard-negative occurrences across 305 records sharing a canonical text identity with another record's gold. Raw exact matching's 293/220 result is the superseded v1 identity rule. First-occurrence metadata filtering reproduces the known source-informed counts of gold loss in 33 records and all-gold loss in 2. The counts are deterministic post-hoc reproductions, not preregistered observations from a

Table 4: FinRank score sensitivity and narrow decision stability.

Model	Tie policy	Hard	Random	Gap
TF-IDF	ties-positive	77.756	91.935	14.179
TF-IDF	strict-win	77.030	90.850	13.821
TF-IDF	half-credit	77.393	91.393	14.000
BM25	ties-positive	74.780	87.789	13.009
BM25	strict-win	74.094	86.760	12.666
BM25	half-credit	74.437	87.275	12.838

wholly frozen application. Reproduction remains a separate construct. In-record MRR/nDCG and all pairwise metrics reproduce exactly. Top-level global ranking metrics differ from the bundled result JSON by up to 0.19 point and metadata-filtered metrics by up to 0.16 point in the declared deterministic environment; a second available environment changes some global values by as much as 0.47 point, while slice aggregates drift by as much as 0.51 point. Without the original score matrices or complete execution environment, the residual global and metadata metric drift cannot be attributed uniquely to tie order or dependency drift. Its cause is unresolved, and the result is a reproducibility boundary rather than a benchmark failure.

6 DISCUSSION AND LIMITATIONS

The two cases support one measurement discipline: visibility, detection and harmfulness concern instruments, predicates and equivalence contracts; score sensitivity, decision stability and reproduction concern metrics, decisions and execution envelopes.

6.1 Scientific limitations

Case 1 uses synthetic perturbations on one public corpus; three seeds cover 119 source-table UIDs but do not estimate a natural failure distribution. Harmfulness is contract-relative: the OCR-like family is unobserved, not proven harmful, and one judge classified its 13 cases as referentially equivalent. The semantic boundary analysis uses one judge on 28 cases. Perfect localization follows explicit stage instrumentation. FinQA does not regenerate downstream result fields, so it establishes a measurement-object boundary, not protocol blindness. FinRank v2 is a post-hoc correction after v1 used the wrong identity rule; its 442/305 and 33/2 counts are not a second frozen application. The audit covers TF-IDF and BM25, not neural or generative systems, and the residual global and metadata drift remains unexplained. MA-CERD, adaptive revalidation and internal AI reviews provide no Paper E evidence.

6.2 Proposed six-step review checklist

The following sequence is a proposed checklist for evaluator authors, not a validated universal method:

1. identify the observation surface and its instruments;
2. bind each perturbation to the recomputed measurement object;
3. inspect score algebra, tie policy and threshold identifiability;
4. preserve source identity and equivalence contracts;
5. test whether the result changes a real report, model-selection or measurement decision;
6. report negative or stable results and the exact reproducibility envelope.

6.3 Integrated interpretation

Case 1 rejects three over-readings: score gaps are not automatically performance gaps, non-response is not automatically harm, and hierarchy alone does not explain localization. Case 2 separates score sensitivity from decision stability and leaves an unresolved reproduction gap. Mutation, observation, score, ranking and claim must refer to the same causally updated object.

7 CONCLUSION

Final correctness cannot show whether a staged evaluation observed, consumed or preserved the relevant evidence. Across two cases we separate visibility, detection, harmfulness, score sensitivity, decision stability and reproduction gap. Case 1 removes score-equivalence, non-response and hierarchy over-readings; Case 2 separates metric sensitivity from decision stability. The result is a bounded audit and proposed checklist, not a universal theory or validated method.

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Generative AI disclosure: OpenAI Codex assisted manuscript organization, English editing and consistency checks. The authors verified claims against cited artifacts. Experimental evidence includes evaluated-system and automated-judge outputs under frozen protocols; deterministic statistics came from supplied scripts. AI did not create human semantic gold or blind review.

8 DATA AVAILABILITY

Case 1 uses public TAT-QA tables; frozen perturbation, checkpoint, aggregation and analysis artifacts are prepared for anonymized release. FinQA is nested in Case 1. Case 2 uses FinRank revision 916dc27e77b90e294ea1b3a2cc5b0abcf07392d1, arXiv 2608.07400v1, FinRank.jsonl, baseline code and results under CC BY-NC 4.0. The wrong-identity v1 application and post-hoc v2 corrections are preserved; raw scores are unchanged. The current private candidate includes the row-level reanalysis input, full-precision and compact FinRank extracts, reproduction scripts and manifests. Historical preregistration, fixtures, application rows and raw score matrices remain in the project evidence store for author-approved release and are not included in this candidate.

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